Rethinking multimodality in enzyme engineering: learning conditional functional landscapes

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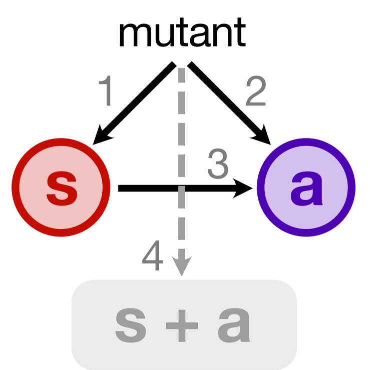
Background

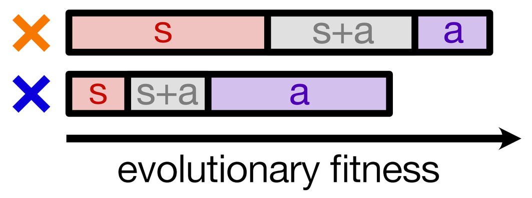
Evolutionary models learn a **convolution of protein functions,** including **stability and activity**

But for **enzyme design**, we want to sample the ***functional*** landscape

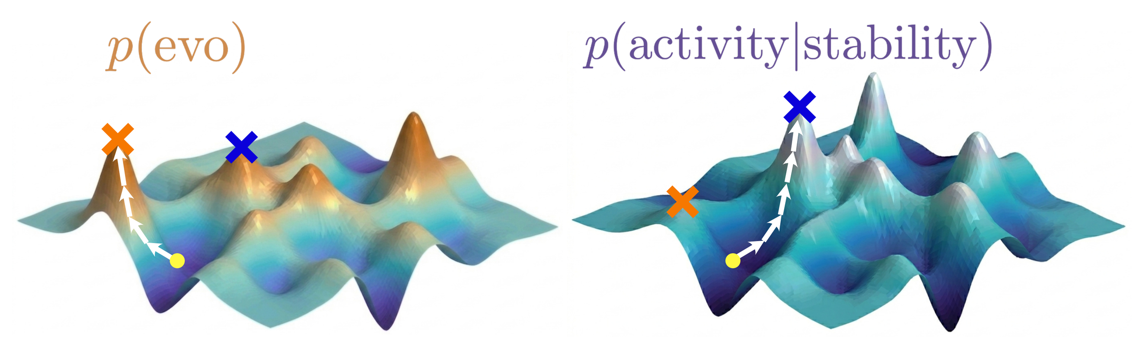
Optimising evolutionary fitness alone favours variants that are active *because* they are stable



1: affects stability
2: affects activity
3: affects activity because of stability
4: joint stability and activity effect



s: stability
a: activity
s+a: joint stability and activity



Evolutionary (left) and conditional activity landscapes (right) for a protein, where a high activity variant (blue cross) is missed if relying on evolution alone.

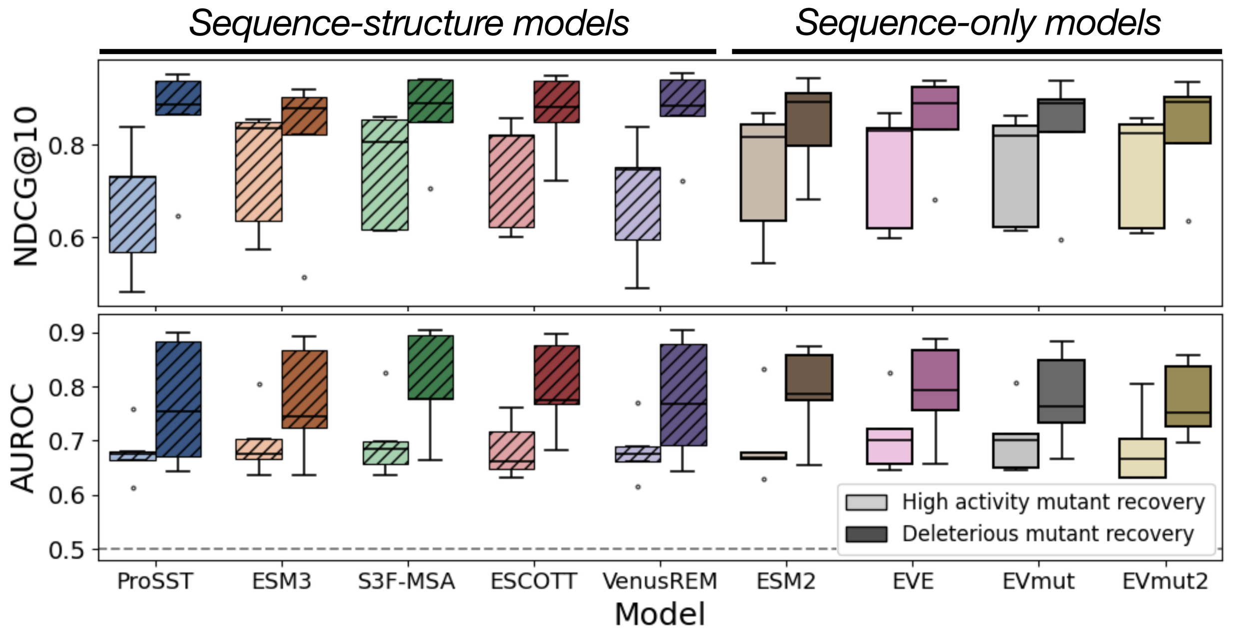
Aim: control for stability effects using structure to learn a ***conditional*** **distribution of protein function**

Multimodal approaches do not learn conditional activity

We used a panel of 5 well-characterised **enzymes with multidimensional deep mutational scans** to inspect the ability of leading multimodal and sequence-based approaches to predict *unconditional* (raw) and *conditional* activity.

How well do models capture *raw* activity?

All models capture the top and bottom 10% of activity similarly

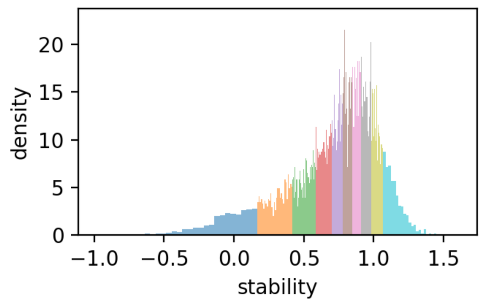


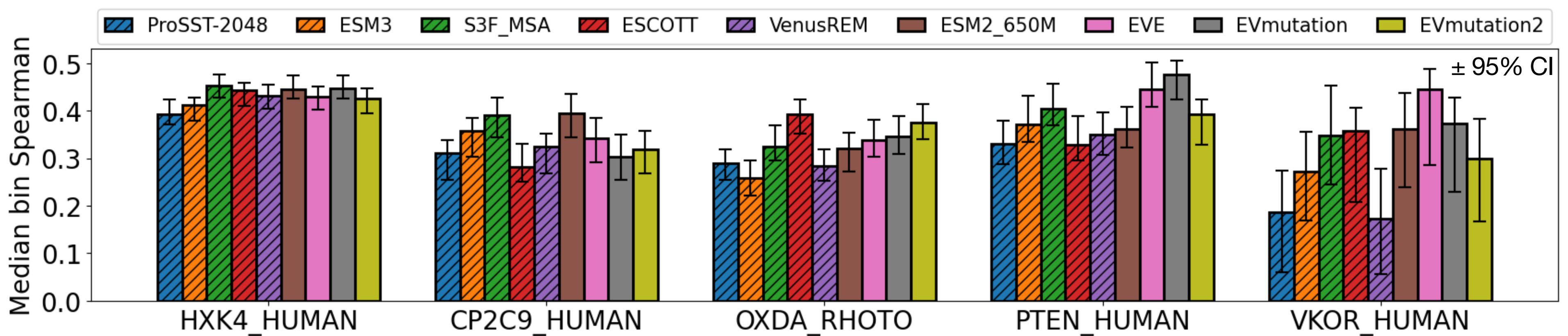
Activity is predicted best when correlated with stability

	Sequence-structure models: ProSST-2048	ESM3	S3F_MSA	ESCOTT	VenusREM	Sequence-only models: ESM2_650M	EVE	EVmutation	EVmutation2	stability
HXK4_HUMAN	0.48	0.48	0.52	0.51	0.51	0.52	0.49	0.51	0.49	0.21
CP2C9_HUMAN	0.67	0.70	0.71	0.65	0.68	0.69	0.66	0.61	0.64	0.75
OXDA_RHOTO	0.36	0.34	0.40	0.44	0.35	0.37	0.39	0.39	0.42	0.25
PTEN_HUMAN	0.51	0.53	0.55	0.51	0.52	0.51	0.59	0.58	0.51	0.32
VKOR_HUMAN	0.30	0.37	0.43	0.41	0.32	0.43	0.43	0.40	0.37	0.25

Spearman correlation coefficients with activity

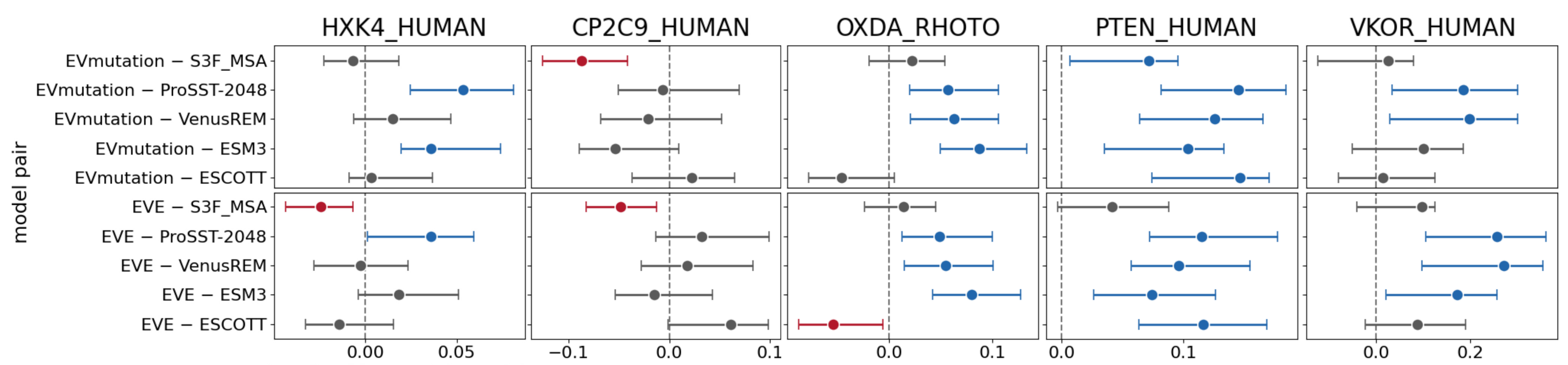
Conditional activity: at similar stability regimes, how well do models recover activity ranks?





Sequence-structure models *do not* learn stability-independent activity effects better than sequence-only methods

No consistent improvement when comparing any leading hybrid model (*e.g.*, S3F-MSA) and evolutionary couplings (EVmutation) or an alignment-based VAE (EVE).

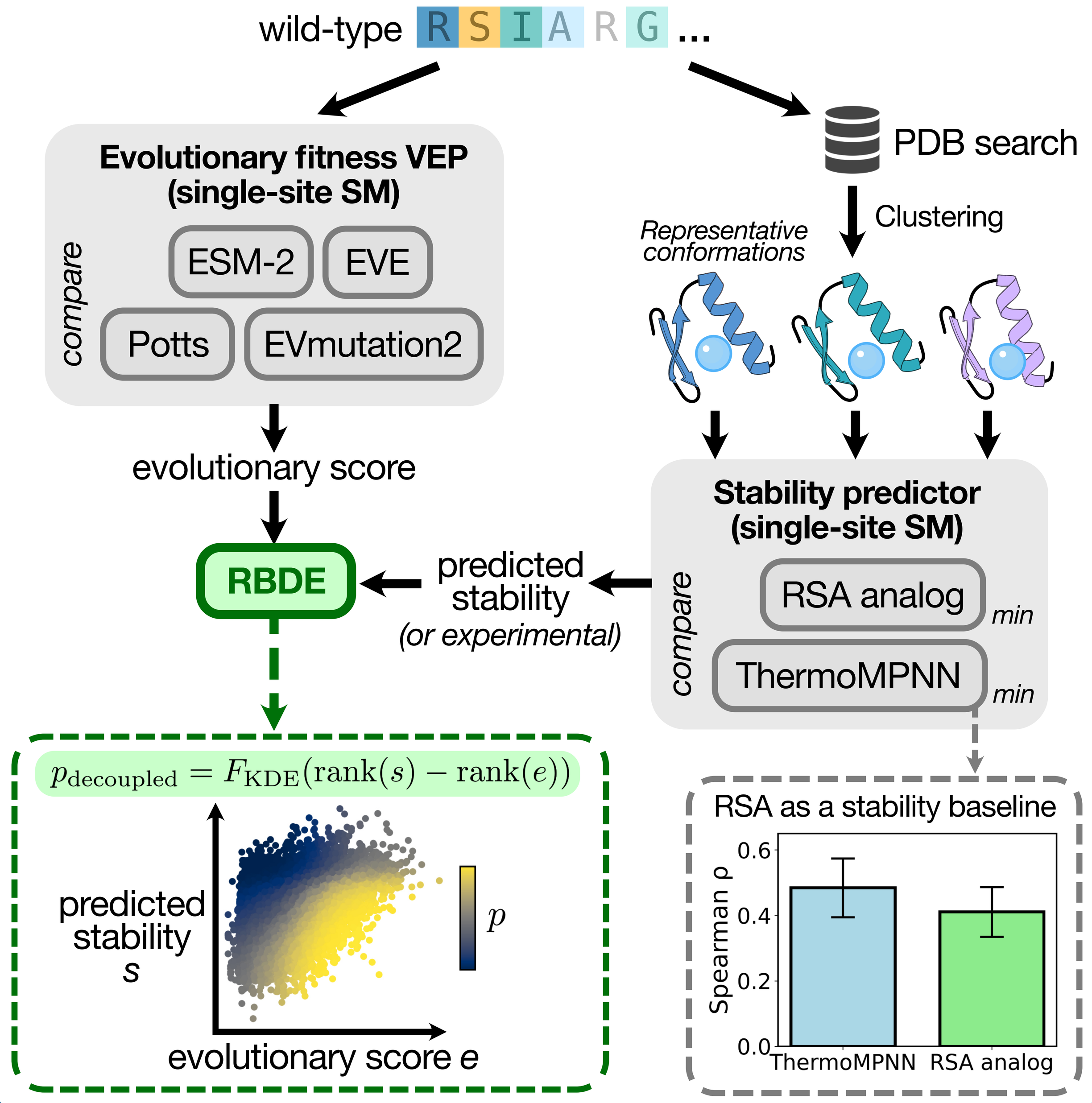


Median bin ΔSpearman with 95% CI after bootstrapping

Evidence for hybrid model improvement · Evidence for sequence model improvement · No difference

Investigating the signal with an independent predictor

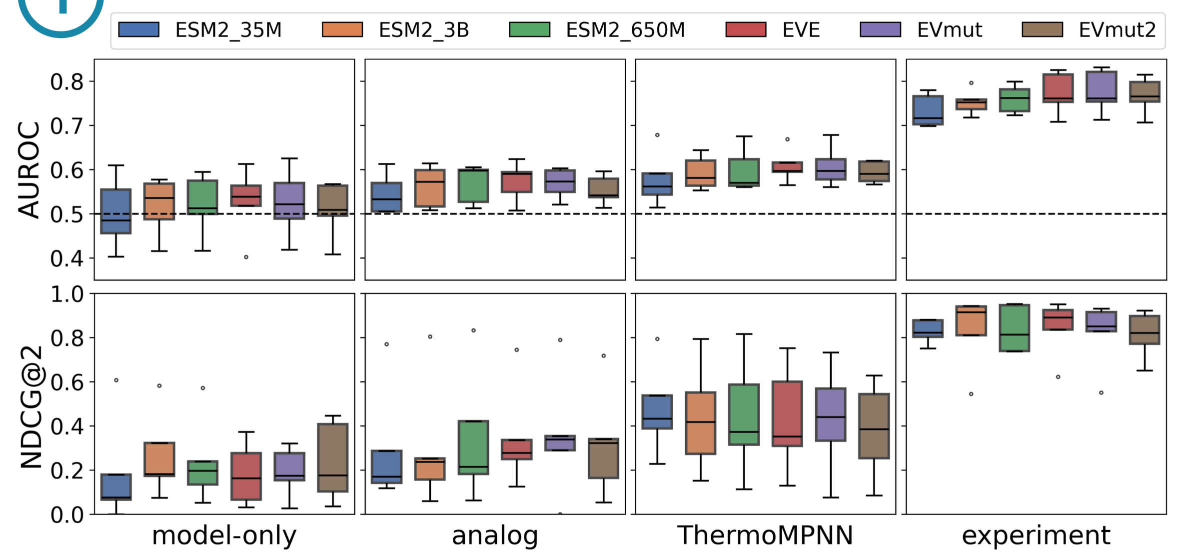
By using **sequence-based evolutionary models** and **stability predictors**, we investigated the *decoupled* signal through building *joint* models via a **rank-based density estimator (RBDE).**



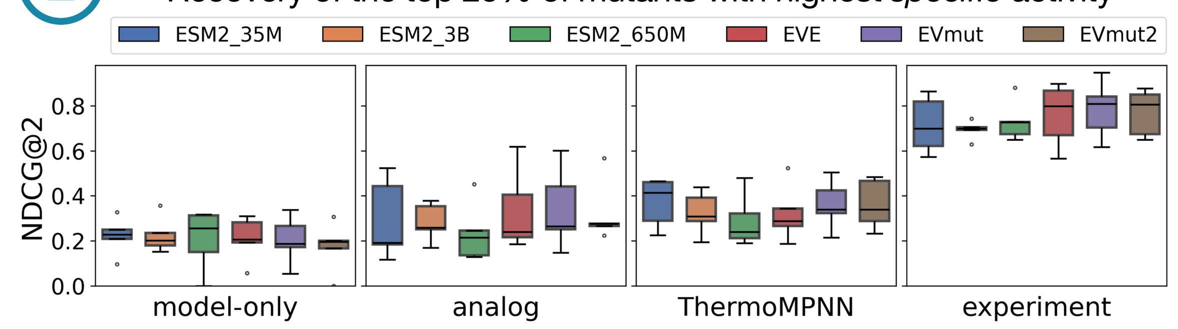
Evaluation: recovery of decoupled activity effects

How well do joint models recover and rank the *most decoupled* mutants?

1. **Stable-but-inactive** as defined by WT ranges
2. High **specific activity** $a_{\text{specific}} = z(a) - z(s)$



Recovery of the top 25% of mutants with highest *specific* activity

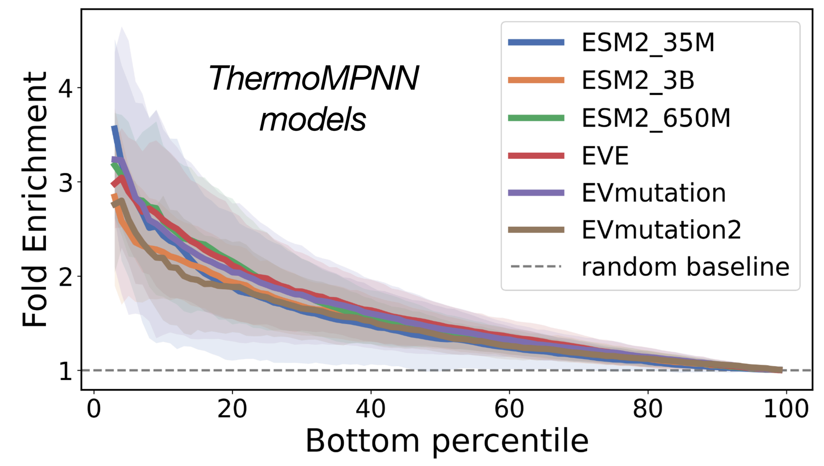


Results using **computational stability predictors differ from experiments**, but appear superior to using evolutionary models alone.

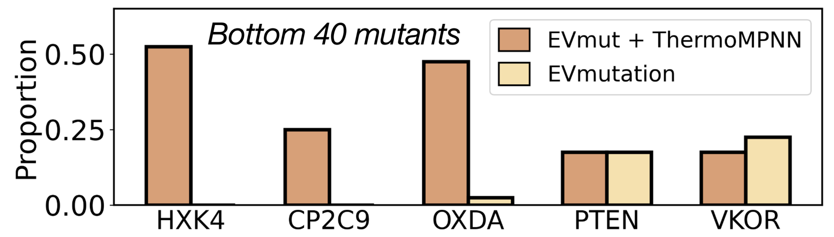
Conditional activity can be used to identify functional sites

Decoupled activity effects from joint models correspond to sites that are **distributed throughout the protein** and are **enriched for ligand binding sites.**

Decoupled scores allow recovering sites with a bound ligand (< 3.5A) throughout the PDB



Joint models identify *stable* mutants in ligand-contacting sites better than VEPs alone



Conclusions

1. Multimodal approaches **do not capture stability-independent enzyme activity** effects better than sequence-only models
2. **Independent predictors can be used to deconvolve activity effects**, but results differ from using experimental stability
3. Conditioning on stability allows **identifying sites that modulate activity** and **bind relevant ligands**
4. Building sequence-structure models with better inductive biases will enable learning functional landscapes for design